

ConnectCIC Metadata Overview

Introduction

The purpose of this document is to describe the ConnectCIC Metadata product. It will cover a general definition of metadata, what ConnectCIC metadata is, and how it can be used in partner applications.

What is metadata?

The general definition of metadata is “data that provides information about other data” and there are many different types. Images may contain metadata that describes the file size, resolution, and the date the image was created, which is an example of descriptive metadata.

ConnectCIC Metadata is a separate XML structure from the ConnectCIC API. Metadata references the API and provides information on how to construct a query or message as well as how to process the incoming response.

What it means for ConnectCIC

Creating forms for CJIS transactions can be complex and time consuming. Keeping them updated with mandated state changes can be even more so. This is compounded when working in more than one state.

ConnectCIC Metadata is an XML document that describes the ConnectCIC API. It contains field definitions (name, length, and values), possible combinations with requirements, and a list of known response types with potential data-mined fields for each transaction in the ConnectCIC developer documentation.

The goal of ConnectCIC Metadata is to provide information that can be used to automate parts of the development process. This saves time whether implementing for a new interface or when dealing with changes to an existing one. It is an extension of the CommSys “write once, use many” philosophy.

How can partners use ConnectCIC Metadata?

ConnectCIC Metadata can be imported into a form or UI builder in partner applications. Partners can develop a provisioning system to import ConnectCIC Metadata and provide a bank of fields and combination sets that map the ConnectCIC transaction name to their software’s form. This allows customization of the UI without having to worry about the back end to ConnectCIC.

When used with ConnectCIC Real Time Data Mining (RTDM), partners can also map parsed return fields with response types listed in Metadata. Parsed fields could then be mapped to

display in formatted layouts or used for importing the data into other locations.

How Metadata is supplied to the Partner

ConnectCIC Metadata is a separate XML file and is not included with the ConnectCIC product. It must be obtained separately for each interface. Partners should contact CommSys for information on Metadata delivery.

Requirements on the Partner

In order to use the ConnectCIC Metadata the partner must develop software to read the metadata file. Metadata is an extension of the ConnectCIC API and doesn't serve any function by itself. Partners will need to create software to consume the Metadata that creates mappings of the contained data to their application. CommSys does not supply a solution to read or provision the Metadata files.

Metadata Example using a Registration Query

Below is an example of a VehicleRegistrationQuery as it would appear in the ConnectCIC Developer Documentation and its corresponding definition in ConnectCIC Metadata. The VehicleRegistrationQuery transaction is designed to return in/out of state registration information, in state hot files, and NCIC hot files when an officer inquires on a vehicle's VIN or license plate data.

An Example Registration Query

Metadata can be a difficult concept to understand. Consequently, we have provided this full example that shows how ConnectCIC Metadata and API tie together to generate a simple inquiry.

Further in the example, we show how a query response is delivered from ConnectCIC.

ConnectCIC API/Developer Documents

The following is the relevant documentation page from the ConnectCIC API guide. This provides a developer with the information necessary to communication with ConnectCIC to execute a lookup on a vehicle by License Plate or Vehicle Identification Number (VIN).

The M/C/O column indicates if the field is Mandatory, Conditional or Optional. Since this transaction can be run by VIN or plate number, those fields are conditional.

VehicleRegistrationQuery

Field Name	XML Tag Name	M/C/O	Size	Possible Values
LicensePlateNumber	LicensePlateNumber	C	10	
LicensePlateTypeCode	LicensePlateTypeCode	C	2	
State	State	O	2	
VehicleIdentificationNumber	VehicleIdentificationNumber	C	30	
VehicleMakeCode	VehicleMakeCode	O	4	

Possible Combinations

1. (In/Out) LicensePlateNumber, [LicensePlateTypeCode, State]

2. (In/Out) VehicleIdentificationNumber, [VehicleMakeCode, State]

Corresponding Metadata

The following is the corresponding metadata.

```
<!--Beginning of a transaction. Defines the ConnectCIC transaction name. -->
<Transaction name="VehicleRegistrationQuery" version="1">
<!--The <Fields> section defines the available fields for the transaction
along with the data type and description. -->
  <Fields>
    <Field name="LicensePlateNumber" type="Alphanumeric" maxLength="10" description=" " />
    <Field name="LicensePlateTypeCode" type="Alphabetic" maxLength="2" description=" " />
    <Field name="State" type="Alphanumeric" maxLength="2" description=" " />
    <Field name="VehicleIdentificationNumber" type="Alphanumeric" maxLength="30" description="" />
    <Field name="VehicleMakeCode" type="Alphanumeric" maxLength="4" description=" " />
  </Fields>
<!--The <Combinations> section defines the available combinations as listed in the
ConnectCIC Developer Documentation with required and optional values. Also lists the
possible return types for the request. -->
  <Combinations>
    <Combination keyReference="IA.QV" primaryFieldReference="LicensePlateNumber">
      <ResponseTypes>
        <ResponseType geoscope="Current" system="MPS" class="NoRecord" />
        <ResponseType geoscope="Current" system="MPS" class="MissingPerson" />
        <ResponseType geoscope="Current" system="MPS" class="Unknown" />
        <ResponseType geoscope="Global" system="NCIC" class="Image" />
        <ResponseType geoscope="Global" system="NCIC" class="ImmigrationViolation" />
        <ResponseType geoscope="Global" system="NCIC" class="MissingPerson" />
        <ResponseType geoscope="Global" system="NCIC" class="NoRecord" />
        <ResponseType geoscope="Global" system="NCIC" class="Parole" />
        <ResponseType geoscope="Global" system="NCIC" class="Probation" />
        <ResponseType geoscope="Global" system="NCIC" class="ProtectionOrder" />
        <ResponseType geoscope="Global" system="NCIC" class="Response" />
        <ResponseType geoscope="Global" system="NCIC" class="SexualOffender" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenArticle" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenBoat" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenGun" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenPlate" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenSecurity" />
        <ResponseType geoscope="Global" system="NCIC" class="StolenVehicle" />
        <ResponseType geoscope="Global" system="NCIC" class="SupervisedRelease" />
        <ResponseType geoscope="Global" system="NCIC" class="ViolentGangOrTerrorist" />
        <ResponseType geoscope="Global" system="NCIC" class="WantedPerson" />
        <ResponseType geoscope="Global" system="NCIC" class="Unknown" />
        <ResponseType geoscope="Global" system="NCIC" class="Unknown" />
        <ResponseType geoscope="Current" system="SVS" class="NoRecord" />
        <ResponseType geoscope="Current" system="SVS" class="StolenVehicle" />
        <ResponseType geoscope="Current" system="SVS" class="Unknown" />
      </ResponseTypes>
      <Requirements>
        <!--Items under <Requirements> <Set> are required fields. -->
        <Set>
          <Field reference="LicensePlateNumber" />
          <!--Items under <Any> are optional. -->
          <Any>
            <Field reference="LicensePlateTypeCode" />
            <Field reference="State" />
          </Any>
        </Set>
      </Requirements>
    </Combination>
  <!--Many transactions will contain multiple combinations as they contain several message keys
that are sent together or under certain conditions. In state vs out of state is one example. -->
  <Combination keyReference="IA.QVK" primaryFieldReference="VehicleIdentificationNumber">
    <ResponseTypes>
      <ResponseType geoscope="Global" system="MPS" class="NoRecord" />
      <ResponseType geoscope="Global" system="MPS" class="MissingPerson" />
      <ResponseType geoscope="Global" system="MPS" class="Unknown" />
      <ResponseType geoscope="Global" system="NCIC" class="Image" />
      <ResponseType geoscope="Global" system="NCIC" class="ImmigrationViolation" />
      <ResponseType geoscope="Global" system="NCIC" class="MissingPerson" />
      <ResponseType geoscope="Global" system="NCIC" class="NoRecord" />
      <ResponseType geoscope="Global" system="NCIC" class="Parole" />
      <ResponseType geoscope="Global" system="NCIC" class="Probation" />
      <ResponseType geoscope="Global" system="NCIC" class="ProtectionOrder" />
      <ResponseType geoscope="Global" system="NCIC" class="Response" />
      <ResponseType geoscope="Global" system="NCIC" class="SexualOffender" />
      <ResponseType geoscope="Global" system="NCIC" class="StolenArticle" />
      <ResponseType geoscope="Global" system="NCIC" class="StolenBoat" />
      <ResponseType geoscope="Global" system="NCIC" class="StolenGun" />
```

```

    <ResponseType geoscope="Global" system="NCIC" class="StolenPlate" />
    <ResponseType geoscope="Global" system="NCIC" class="StolenSecurity" />
    <ResponseType geoscope="Global" system="NCIC" class="StolenVehicle" />
    <ResponseType geoscope="Global" system="NCIC" class="SupervisedRelease" />
    <ResponseType geoscope="Global" system="NCIC" class="ViolentGangOrTerrorist" />
    <ResponseType geoscope="Global" system="NCIC" class="WantedPerson" />
    <ResponseType geoscope="Global" system="NCIC" class="Unknown" />
    <ResponseType geoscope="Global" system="Unknown" class="Unknown" />
    <ResponseType geoscope="Global" system="SVS" class="NoRecord" />
    <ResponseType geoscope="Global" system="SVS" class="StolenVehicle" />
    <ResponseType geoscope="Global" system="SVS" class="Unknown" />
  </ResponseTypes>
  <Requirements>
    <Set>
      <Field reference="VehicleIdentificationNumber" />
      <Any>
        <Field reference="VehicleMakeCode" />
        <Field reference="State" />
      </Any>
    </Set>
  </Requirements>
</Combination>
<Combinations>
</Transaction>

<!-- Response Types detail the list of fields that can potentially be returned with that System and Class tag set.
These can be used in conjunction with ConnectCIC RTDM. -->
<ResponseType system="SVS" class="StolenVehicle" description="Stolen Vehicle System" version="2">
  <Components>
    <Component name="FreeText" type="Text" />
    <Component name="LicensePlateNumber" type="Text" />
    <Component name="LicensePlateStateCode" type="Text" />
    <Component name="LicensePlateTypeCode" type="Text" />
    <Component name="LicensePlateYear" type="Year" />
    <Component name="NormalizedLicensePlateYear" type="Year" />
    <Component name="OriginatingAgencyCaseNumber" type="Text" />
    <Component name="OriginatingAgencyIdentifier" type="Text" />
    <Component name="TheftDate" type="Date" />
    <Component name="VehicleIdentificationNumber" type="Text" />
  </Components>
</ResponseType>

```

1. Transaction Definition

The Transaction definition primarily defines the name and potentially the version number of the transaction. ConnectCIC has the potential for multiple versions of the same transaction to exist simultaneously. To resolve the proper version, the Metadata provides a version number in the Transaction Definition section.

In this example we are using the VehicleRegistrationQuery.

2. Field Definition

The <Field> section will contain the available fields with their XML tags, data type, length, a description of the field, and component fields if available. These will match the fields listed in the ConnectCIC API documentation for the corresponding transaction.

3. Transaction Combinations

The <Combinations> section will detail the different ways the transaction can be run. These should reflect the Possible Combinations section in the API documentation. Each combination will be defined by the message key and primary field. In the above example, this transaction can be run by the License Plate combination or the VIN combination.

4. Combination Requirements

Each <Combination> section will also contain a <Requirements> section that details the minimum fields that must be submitted for the transaction to run along with any conditional or optional fields available. Fields listed under <Requirements><Set> are required and

anything under <Any> is optional. These should match the API document for the Possible Combinations.

5. Response Information

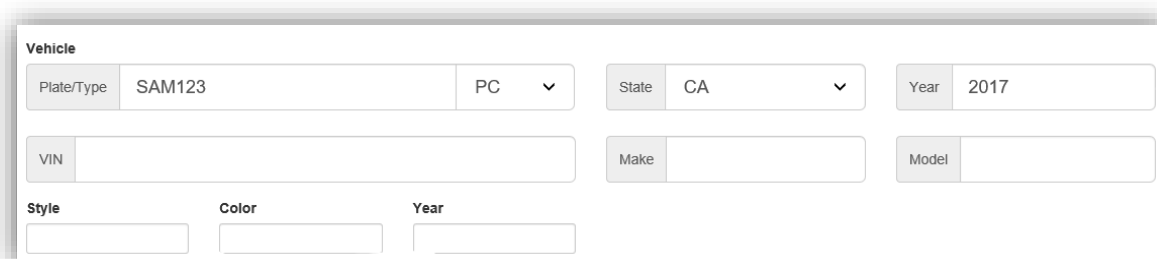
Each combination will list the possible responses that can be returned. These Response Types will be defined by the <System> and <Class> tags which are returned by ConnectCIC, highlighted yellow in the example below. The <System> tag will define the database from which the response was generated. For a vehicle registration response, this would likely be “DMV”. In this example it is the California Stolen Vehicle System (SVS). The <Class> tag will define the type of response. In this case, it is Stolen Vehicle.

Response Types also describe the fields that can be potentially returned and parsed with the ConnectCIC Real Time Data Mining (RTDM) feature. If the RTDM feature is enabled, ConnectCIC will return individual data elements following the response that can be used by the partner software, highlighted in gray below. ConnectCIC Metadata will list these fields for each transaction.

Provisioning the Partners Forms

Idealized User Form

Below is an example of what the form for the `VehicleRegistrationQuery` might look like in a partner application. This is not something ConnectCIC presents but is what the partner application presents to the end user.



Partial Example Response for Vehicle Registration

The following is what a response from ConnectCIC to the partner software will look like. The block data will be enclosed in the <MessageData> element and the parsed information will follow. These XML tags will correspond to the assigned Response Type in the ConnectCIC Metadata beginning with the <System> and <Class> tags for identification.

```
<MessageData>4CAD1000055.IA
QV.CA1234567.LIC/SAM123
```

```
INQUIRY MATCH ON LIC/SAM123
STOLEN VEHICLE
```

```
LIC/SAM123 LIS/CA LIY/2017 LIT/PC 2017
CHEV BTM PU WHI VIN/VEHICLEIDNUMBER
ORI/CA1234567 OCA/16123456 FCN/1234124124515 DOT/20161231
NIC/V123456788
MIS/TWO PLATES, ALL KEYS ACCOUNTED FOR, MISSING THE C IN CHEVROLET
```

NOA/N ENT/ON CALIF AND NCIC FILES
IMMEDIATELY CONFIRM WITH ORI/CA1234567 MODESTO PD
MNE/MNE1 TELEPHONE 999 888-7777

* * * * * VICTIM DATA * * * * *

VNM/XAULT, TIFFANY
VAD/PO BOX 999999
VCY/SACRAMENTO VSA/CA VZP/54321
VDP/(988) 887-6543 VNP/(988) 887-6542
ADB/19550101

```
</MessageData>  
  <System>SVS</System>  
  <Class>StolenVehicle</Class>  
  <TheftDate>20161231</TheftDate>  
  <LicensePlateNumber>SAM123</LicensePlateNumber>  
  <LicensePlateStateCode>CA</LicensePlateStateCode>  
  <LicensePlateTypeCode>PC</LicensePlateTypeCode>  
  <LicensePlateYear>2017</LicensePlateYear>  
  <FreeText>TWO PLATES, ALL KEYS ACCOUNTED FOR, MISSING THE C IN CHEVROLET </FreeText>  
  <OriginatingAgencyCaseNumber>16123456</OriginatingAgencyCaseNumber>  
  <OriginatingAgencyIdentifier>CA1234567</OriginatingAgencyIdentifier>  
  <VehicleIdentificationNumber>VEHICLEIDNUMBER</VehicleIdentificationNumber>  
  <Broadcast>1</Broadcast>  
  <IntTransId>00004Y</IntTransId>  
  <MessageType>SVS</MessageType>  
</Response>  
</Transaction>  
</LawEnforcementTransaction>
```

The highlighted data above is generated by the Real Time Data Mining (RTDM) feature of ConnectCIC. The <System> and <Class> tags, highlighted in yellow, identify the source of the return and the type of return. These are generated by ConnectCIC and correspond to the ResponseType defined in the ConnectCIC Metadata. The data highlighted in gray is parsed from the return into individual tags for consumption in the partner application. These are also defined by the ResponseType in the ConnectCIC Metadata.